

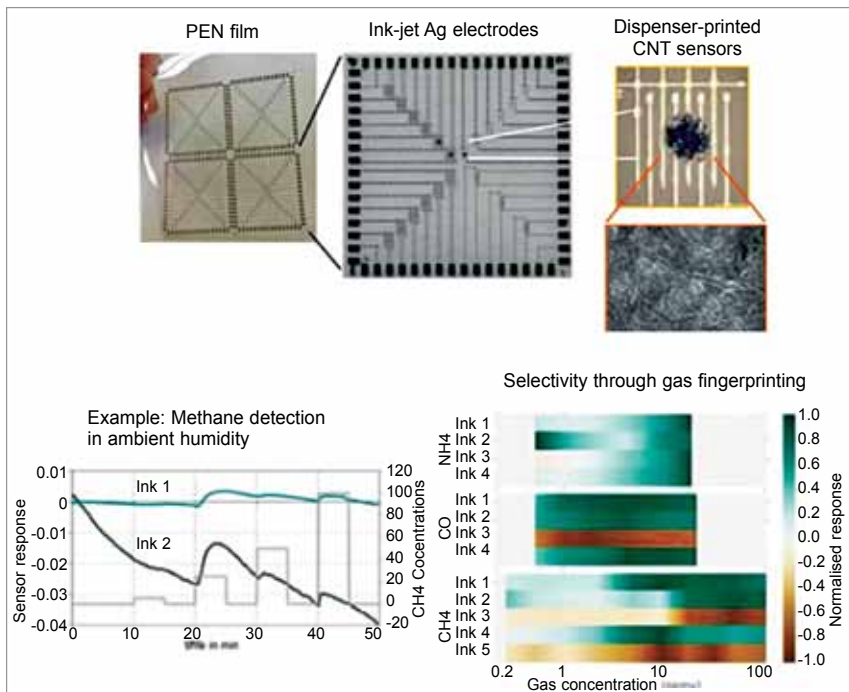


Fig. 8: Electronic nose for gas fingerprinting (Credit: Xerox PARC)

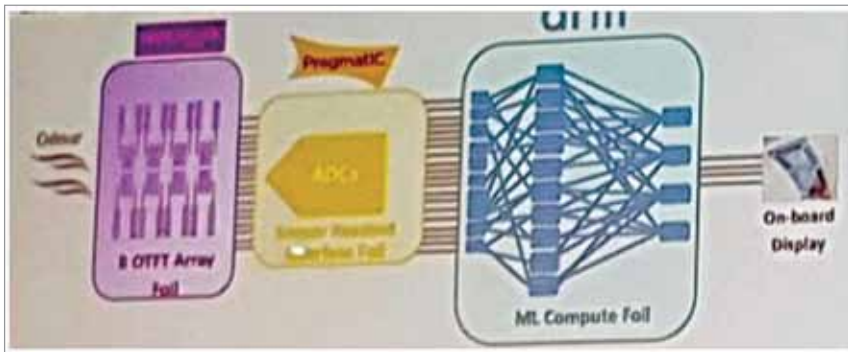


Fig. 9: Electronic nose with FHE (Credit: PragmatIC)

and sent to an app for providing efficient care.

CareWear and Atlantic Therapeutics provide light-emitting skin patches that monitor the human body's healing process, and pants that have embedded electrodes for improving thigh muscle strength.

Smart packaging and IoT/IoT.

Wiliot has released flexible, wireless RFID tags that have a silicon IC and an antenna for harvesting energy from electromagnetic radiation. It can also perform sensing by analysing a metal's conductivity. Applications include supply-chain tracking and cloud sensing.

Boeing's prototype sensor for smart factory or Industry 4.0 incorporates a thin, flexible IC that offers numerous functionalities for monitoring temperature, strain, humidity, force applied, etc on industrial equipment and aircraft parts.

Thanks to the large investment by PragmatIC, a UK based company that produces flexible metal-oxide ICs, it is now possible to inexpensively manufacture such ICs on a large scale for enhanced RFID functionalities. The Swedish firm, Ynvisible, has partnered with Evonik and Epishine to develop self-powered shelf-level packaging

signage along with digital item price indicators.

A unique capacitive sensor made by Bare Conductive from a low-cost carbon based ink changes its capacitance with a rise or fall in water level. This information is then sent to the cloud for processing.

Automotive. PolyTC has successfully used its transparent functional films in some of Volkswagen's and Audi's latest models. The transparent capacitive film consists of a metal mesh that can detect changes in capacitance when in contact with different areas.

Printed conductors for heating is another attractive prospect. Instead of inefficiently heating the entire vehicle interior, the printed heaters focus on heating only specific areas near the occupants such as doors, windows, and seats - thus reducing the total required heating energy by 20%. This technique has been tested out in Volkswagen and BMW models.

Ideal for the future

After going through all the above innovations, we can conclude that the massive growth of IoT will surely increase the demand for sensors. As of now, printed flexible electronics can greatly fulfil this cost-effectively and with few materials.

The year 2021 has been quite encouraging for printed electronics companies to raise funds for applications and address the needs of large markets. This is expected to continue in 2022, thus leading to a wider adoption in areas where processing is still done using conventional electronics. **EFY**

The article is based on the presentation 'Commercially Compelling Printed/Flexible Sensor Innovations' at IDTechEx and the market research report 'Printed and Flexible Sensors 2021-2031.' The article was compiled by Vinay Prabhakar Minj, Technology Journalist at EFY.